

Task Manager App

A calendar-centric life dashboard: tasks, fitness, medication, payments, and expenses in one place



What's new in v3:

- User / Admin / Caregiver role architecture
- Offline handling & real-time offline detection
- Timezone handling for reminders and escalation
- Full revocation & teardown model, incl. autonomy safeguards
- Caregiver reassignment flow
- Two lifecycle flowcharts

1. Core concept

The app is a calendar-centric task manager. The **Home screen is a live calendar**, and four separate life-domain modules feed into it: **Fitness**, **Reminders** (Pill, Payment, Appointment, Custom), and **Expense**. The calendar is strictly an **aggregator and preview layer** — it never owns or duplicates module data.

2. Navigation structure

Tab	Role
Home	Calendar view. Aggregates and previews entries from all other modules.
Fitness	Workout logs, goals, and AI-assisted plan generation.
Reminders	Pill, Payment, Appointment, and Custom — grouped under one tab, each with its own icon.
Expense	Categorized spending ledger; rolls up into daily, weekly, and monthly totals.

3. The "+" add flow

Module-first, always. Global "+" shows module picker, then that module's categories. Tapping a specific date on Home shows module options directly with the date pre-filled.

4. Reminders module

4.1 Pill MVP: manual entry v2: auto-lookup

Functions as a lightweight medical ID for elderly users. **Fields:** name, dosage, time(s)/day, expiry, photo.

4.2 Payment / Bill

Fields: payee, amount, due date, recurring flag. Links into Expense — see Section 6.3.

4.3 Appointment

Kept deliberately simple — title, location, time only.

4.4 Custom

User-picked emoji represents a freeform task. Same-day reuse of an emoji groups tasks under that bucket (distinct timestamps required) rather than duplicating entries; the calendar shows a count badge. Tracking is set per item as streak/checklist or numeric.

4.5 Timing & recurrence

Timing is module-specific (Pill = times/day, Payment = days-before-due, Fitness/Custom = simple time). Recurrence offers quick presets (next day, 2/3 days after, same day next week/month) plus full custom flexibility for irregular patterns.

4.6 Missed / overdue — escalation ladder

Loud by default, with a per-user toggle to quiet it down. Missed pills specifically stay visible to a linked caregiver.

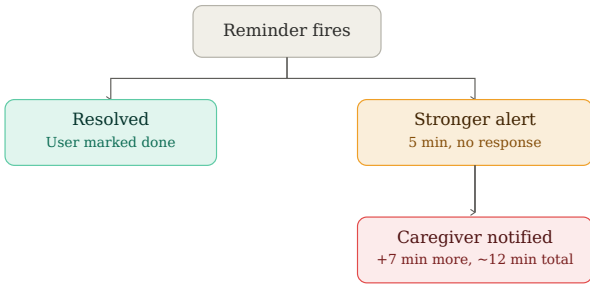


Fig 1 — Missed reminder escalation flow

5. Fitness module

Objects: Workout log (retrospective) and Goal (prospective). Goals drive the calendar forward — planned days appear in advance.

Plan creation, two paths: an AI-assisted chatbox that gathers fitness level/goal/equipment/days and proactively drafts a plan (user refines after), or fully manual entry. Diet-plan generation was considered and removed — workouts only.

Plan vs. actual: once a user edits an AI-suggested workout, the edited version becomes the real plan. Actual logged results are compared against that edited plan for goal/streak tracking.

v2 Wearable / step-counter sync — MVP is manual logging only.

6. Expense module

Categories: Food, Transport, Bills/Utilities, Shopping, Health, Other. Tapping a date shows that day's spend; an end-of-month section shows weekly breakdowns.

Payment ↔ Expense link: a paid Payment auto-creates a linked Expense entry (Bills), with a per-payment opt-out toggle. Paid and unpaid both count toward the monthly total, visually distinguished.

7. Home calendar — visual system

Month view is the default. Dates are shown as density-coded shapes, not icons:

Task count	Color	Edge style
0	Translucent / neutral	—
1-4	Green	Soft, rounded
5-7	Orange	Medium
8+	Red	Rough, sharper

Edge shape is a deliberate accessibility layer alongside color. No numbers or icons live inside grid cells — daily/monthly spend lives behind tap-in detail and the end-of-month summary section.

8. Onboarding & signup

Signup collects name, age, location permission, phone, and email, with a separate setup path for "for a child" or "for an elderly person." New users are forced through **Pill + Payment** (fixed, highest-consequence) before a skip option appears; which modules are forced can vary by profile risk level, and elderly/45+ profiles are forced through all 4. The walkthrough offers real data entry or a pre-recorded demo video.

9. App architecture — User / Admin / Caregiver roles

Role	Definition
User	The person themselves — an elderly person using their own account and a regular user are the same role.
Admin	The role held by someone overseeing another person's account. Holds overall authority — including assigning and reassigning who acts as Caregiver.
Caregiver	A role the Admin assigns (to themselves or to someone else, e.g. a hired nurse) specifically for day-to-day monitoring.

Open: Admin holds overall authority, but the User can take back control in certain ways when needed — the exact scope of this override is not yet fully specified.

10. Caregiver monitoring

10.1 Setup & visibility

- Caregiver access requires **mutual consent** from both people to begin — never unilateral.
- The caregiver sees **full detail** (medication names, dosages, expiry), not just a status signal.
- The monitored person can always see that they are being watched — monitoring is never silent.

10.2 Revocation & teardown

- The **caregiver** can revoke access at any time.
- The **elderly/monitored person** can also revoke access independently — either via a 6-digit code sent to the caregiver's phone (the mutual/routine path), or by stating a reason in writing via mail/email (the one-sided path, usable without the caregiver's cooperation).
- A child profile above 18 can revoke access themselves; a child profile below 18 cannot revoke access once granted.
- When access is revoked, it ends **immediately** and all data is wiped from the former caregiver's view — no lingering historical access.

Autonomy safeguard: giving the monitored person an independent, caregiver-cooperation-free revocation path (the written-reason route) prevents monitoring from becoming a one-way, permanent loss of control — a real risk this design deliberately closes off.

10.3 Caregiver reassignment

If a caregiver quits, the Admin can assign a new one. Historical data is preserved through the transition rather than lost.

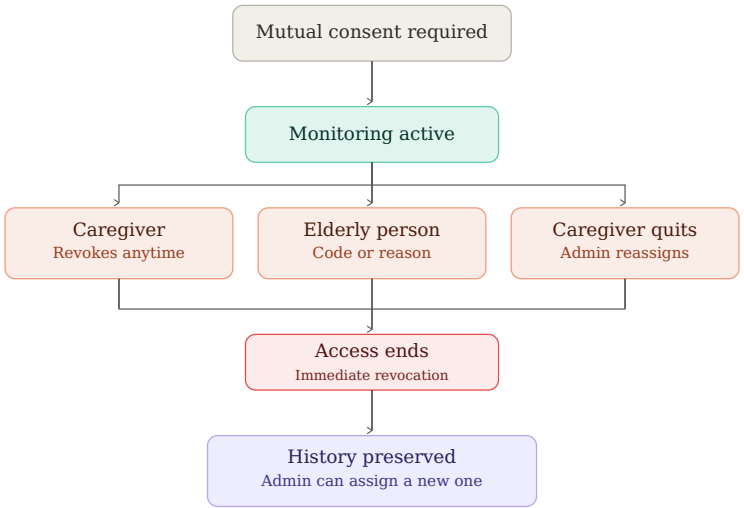


Fig 2 — Caregiver access lifecycle

11. Offline & connectivity handling

- Reminders fire **locally** on the monitored person's phone even with no network — the schedule doesn't depend on connectivity.
- Offline detection is **real-time**: the moment a monitored phone goes offline, the caregiver is notified on the spot — not retroactively after reconnection.
- The caregiver can see scheduled/upcoming reminders in real time (pre-known data), but cannot see completion status (done / not done) until connectivity is restored, at which point all queued data syncs at once.

12. Timezone handling

Reminders and escalation timing run entirely on the **monitored person's own timezone** — never the caregiver's. The caregiver's app only needs to display timestamps intelligibly across timezones; the underlying schedule and escalation clock are unaffected by where the caregiver is.

13. Open risks & unresolved items

- User override scope.** Admin holds overall account authority, but the User can reportedly take back control in certain ways — exact mechanics not yet defined.
- Elderly-simplicity vs. system depth.** Needs a UX pass once real screens exist to confirm the simple path (add a pill, view calendar, get reminded) never requires touching AI plans, emoji grouping, or multi-role account management.
- Notification delivery engine.** Escalation timing is defined; the underlying push infrastructure to deliver it reliably is still a build item.

14. Deferred / v2 scope

Item	Why deferred
Pill drug-database auto-lookup	Requires integrating an external medicine database.
Fitness wearable / step-counter sync	Real integration project (device APIs).
Full multi-role permission system	Core rules are defined (Section 9-10); the complete permission engine is bigger than a first release.

15. Still to design

- User-override mechanics (Section 13).
- Visual mockups for each module's detail screens.
- Data security / encryption approach (engineering-level, not IA, but must not be forgotten given the medical and financial data involved).